

The Systematic Risk of Idiosyncratic Volatility

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Abstract

We show that the pricing of idiosyncratic volatility (IV) is due to unaccounted systematic risk, which affects a large number of asset pricing anomalies. A single common IV component explains one third of variation in IV. Mispricing arises when sorting stocks by the part of IV predicted by exposure to the common IV component, but not when sorting by residual IV. We propose the return difference between stocks with high and low predicted IV, *PIV*, as a proxy for the missing risk factor. *PIV* is significantly related to the common components of the returns of several important anomaly-based trading strategies. Controlling for *PIV* helps explain the earnings surprise, net stock issues, profitability, as well as illiquidity anomalies, and completely explains the financial distress anomaly. Overall our results suggest that the IV puzzle reflects systematic risk related to the average stock volatility and credit risk.

1 Introduction

A plethora of expected return strategies that are not explained by the market, size, and book-to-market factors exist in the asset pricing literature (see Cochrane (2011)). Among these, one that has received considerable attention, is the idiosyncratic volatility (IV) puzzle. Since Ang, Hodrick, Xing and Zhang (2006) initially pointed out the puzzling underperformance of stocks with high idiosyncratic return volatility, many attempts have been made at finding an explanation.¹ Confronted with a robust phenomenon and without a “simple explanation,” Ang, Hodrick, Xing and Zhang (2009) conclude that a missing risk factor is the most likely explanation, but that further research is needed. In this article, we explore the commonality of idiosyncratic volatility in the cross-section of stocks. We hypothesize that exposure of a stock’s IV to a common component of idiosyncratic volatility reflects exposure to systematic risk, which is not captured by the Fama and French (1993) three factor model. We show that this exposure indeed explains the IV puzzle and helps explain a large number of other existing return anomalies. Furthermore, we document that the common IV component is related to both the average stock volatility and the default spread.

If existing asset pricing models such as the Fama and French (1993) three factor model do not fully account for all risk factors, stocks with higher exposure to missing factors will likely have higher (seemingly) idiosyncratic return volatility than stocks with no exposure to missing factors. Sorting stocks on idiosyncratic volatility would then reveal return differences reflecting differences in exposure to unaccounted risk factors. Missing risk factors would induce commonality in IV and stocks with high exposure to common IV components could serve as a proxy for the missing risk factors. Finally, systematic but so far unaccounted risk factors behind the IV puzzle should matter for asset pricing in general and in particular for other asset pricing puzzles.

We provide evidence that is consistent with these conjectures: First, we show that monthly idiosyncratic return volatilities commove substantially. A single common com-

¹The finding by Ang et al. (2006) is particularly puzzling as earlier research (see, in particular, Lehmann (1990)) had been consistent with predictions, for example by Merton (1987), that stocks with higher residual variation earn higher expected returns.

ponent explains 32% of the variation in idiosyncratic volatility. We then decompose idiosyncratic volatility into predicted (by the common component(s)) idiosyncratic volatility and residual idiosyncratic volatility. Almost all of the return differential documented by Ang et al. (2006) occurs in the portfolios that are sorted by their predicted idiosyncratic volatility. Almost no return differential exists when stocks are sorted by their residual idiosyncratic volatility.

Second, we show that the first common idiosyncratic volatility component is significantly correlated with business cycle variables such as the default spread even after controlling for the average stock volatility (e.g. Chen and Petkova (2012)). Particularly, increases in the default spread are associated with increases in the common component of idiosyncratic volatility.

Third, we repeat the decomposition of idiosyncratic volatility out-of-sample and construct a new risk factor, *PIV*, which is the return on a portfolio of the stocks in the highest quintile of *predicted* idiosyncratic volatility minus the return on a portfolio of the stocks in the lowest quintile of *predicted* idiosyncratic volatility. Formal asset pricing tests confirm that *PIV* plays an important role in explaining excess returns of the 30 Fama-French industry portfolios and that *PIV* is priced, even in the presence additional risk factors such as a momentum and a liquidity factor.

Forth, we show that *PIV* helps explain other prominent return anomalies. Specifically, we examine the return anomalies related to earnings surprises, financial distress, net stock issues, profitability, and illiquidity, which cannot be explained by the three Fama-French factors together with the momentum and the Pastor-Stambaugh liquidity factors. Among all these anomalies, the financial distress anomaly is particularly affected by *PIV*. Indeed, the alpha of this strategy decreases by about two thirds once we include *PIV* as a factor. Importantly, across all anomalies, the average absolute correlation between the residual anomaly returns drops from 30% when only including the five risk factors above to 14% when *PIV* is included as an additional factor. Our paper thereby contributes to a broader research agenda laid out by Cochrane (2011) and aimed at reconciling a large set of existing anomalies.

Finally, we provide several robustness checks. First, we show that our findings hold when

using idiosyncratic volatility estimated from quote-midpoint prices as proposed by Han and Lesmond (2011). Second, we confirm that *PIV* is priced when test assets exclude low credit quality firms (Avramov, Chordia, Jostova and Philipov (2013)). Finally, we document that *PIV* matters for asset pricing even in the presence of factors that mimic innovations in average variance and in average correlation (Chen and Petkova (2012)).

We contribute to the literature on the idiosyncratic volatility puzzle by providing evidence that the puzzle is related to a missing risk factor related to both the average stock volatility and to financial distress. We thereby reconcile the apparently contradictory explanations for the IV puzzle of Chen and Petkova (2012) and Avramov et al. (2013).² While Chen and Petkova (2012) propose innovations in average equity return variance (one of two components of market variance, the other being average equity return correlation) as an omitted risk factor to explain the IV puzzle, Avramov et al. (2013) argue that the IV puzzle is limited to firms with low credit quality. Specifically, Chen and Petkova (2012) show that high IV firms have high research and development expenditures and they attribute the negative premium of high IV stocks to the fact that due to their growth options they serve as a hedge against increases in the average equity return variance. Avramov et al. (2013) on the other hand argue that when a firm enters into distress, its idiosyncratic volatility increases, and at same time a negative stock return is realized due to the expected costs of distress and credit downgrades. The negative return of high IV stocks therefore reflects a characteristic of low credit quality firms, not the exposure to a systematic factor. We find that as in Chen and Petkova (2012) the IV puzzle is due to a missing risk factor, not a characteristic of low credit quality firms as suggested by Avramov et al. (2013). As a matter of fact, *PIV* is an important asset pricing factor even when the test assets exclude all firms that Avramov et al. (2013) identify as of low credit quality. But we also find that the common IV component is significantly related to the default spread in addition to the average equity

²Other papers have suggested that the findings of Ang et al. (2006) are due to the choice of frequency (Huang, Liu, Rhee and Zhang (2010)), weighting (Bali and Cakici (2008)), illiquidity (Han and Lesmond (2011)), or the specific measurement of volatility (Fu (2009)). Baker and Wurgler (2006) find that the relationship between idiosyncratic volatility and expected returns depends on investor sentiment. Zhang (2006) and Jiang, Xu and Yao (2009) confirm the findings by Ang et al. (2006) and provide an information based explanation for the lower returns of high idiosyncratic volatility stocks. Ang et al. (2009) subject their earlier finding to a long list of robustness checks, considering possible confounding factors such as firm characteristics, liquidity, private information, and skewness as well as option pricing explanations.

return variance. Furthermore, we document that *PIV* is particularly important in resolving the financial distress anomaly. Naturally, the relationship between the IV puzzle and the financial distress anomaly that we document does not explain the puzzling result that firms in financial distress have low abnormal expected return (e.g. Dichev (1998) and Campbell, Hilscher and Szilagyi (2008)). Our findings, however, indicate that our common volatility factor, *PIV*, captures both the IV puzzle and the financial distress puzzle.

Our study is also related to a still growing literature investigating the determinants of idiosyncratic volatility. Campbell, Lettau, Malkiel and Xu (2001) show that the average idiosyncratic volatility across the universe of U.S. stocks is not constant over time, suggesting that economy-wide factors affect the level of idiosyncratic volatility.³ We contribute to this literature by explicitly linking commonality in idiosyncratic volatility to business cycle variation, in particular in form of variation in the average equity variance and in the default spread.⁴ Connor, Korajczyk and Linton (2006) model idiosyncratic volatility as a function of macroeconomic and firm-specific latent factors. Differently from our study, though, they do not examine firm-level volatility that is unexplained by the three Fama-French factors nor do they explore the asset pricing implications of their findings.

The rest of this article is organized as follows: Section 2 presents evidence that the idiosyncratic volatility is related to a missing factor in the Fama and French (1993) model. Section 3 discusses the macro-economic content of the first common IV factor. Section 4 introduces our new risk factor, *PIV*, and shows its relevance for asset pricing. Section 5 investigates whether *PIV* contributes to resolving other existing return anomalies and Section 6 provides several robustness tests. Section 7 concludes.

³Several authors have proposed explanations for these variations, mainly focusing on changes of the composition of listed firms (see, for example, Xu and Malkiel (2003), Brown and Kapadia (2007), and Fink, Fink, Grullon and Weston (2010)) or on factors that ultimately affect cash flow variability, for example, reflecting changing growth options, research and development expenditure, or product market competition (e.g. Zhang (2010), Cao, Simin and Zhao (2008), Comin and Philippon (2005), Gaspar (2006), and Irvine and Pontiff (2009)). Others point to changes in market efficiency (Chun, Kim, Morck and Yeung (2008)), institutional ownership (Bennett, Sias and Starks (2003)) or speculative behavior (Brandt, Brav, Graham and Kumar (2010)).

⁴Bekaert, Hodrick and Zhang (2012) examine average idiosyncratic volatilities across a large set of countries and also find evidence that average idiosyncratic volatilities co-move with business cycle risk.

2 Idiosyncratic volatility and a missing factor

In this section, we show that the idiosyncratic volatility finding of Ang et al. (2006) is consistent with one or more risk factors missing from the Fama and French (1993) model. We start, in Section 2.1, by replicating the idiosyncratic volatility puzzle of Ang et al. (2006). Then, in Section 2.2, we extract common components from stocks' idiosyncratic volatilities and use them to decompose idiosyncratic volatility into a predicted component as well as a residual component. We then sort portfolios based on stocks' predicted and residual idiosyncratic volatilities and show that the predicted component of idiosyncratic volatility is priced while the residual component is not.

2.1 Idiosyncratic volatility puzzle of Ang et al. (2006)

We obtain daily returns for all stocks in the CRSP data base since July 1963. For a stock to be included in our data set, we require at least 60 months with at least 15 daily returns per month. To ensure comparability between our analysis and the results presented in Ang et al. (2006), we limit our data set to observations between 1963 and 2000. Our final data set consists of 8,274 stocks with over 27 million daily return observations.

As in Ang et al. (2006), we perform time-series regressions at the daily frequency (t) separately for every month (m) of daily returns ($r_{i,m,t}^e$) in excess of the risk-free rate onto daily returns on the three Fama-French factors: market (MKT), size (SMB), and value (HML):

$$r_{i,m,t}^e = \alpha_{i,m} + \beta_{i,m}^{MKT} MKT_{m,t} + \beta_{i,m}^{SMB} SMB_{m,t} + \beta_{i,m}^{HML} HML_{m,t} + \varepsilon_{i,m,t}. \quad (1)$$

We measure stock i 's idiosyncratic return volatility in month m , $IV_{i,m}$, as the square root of the average squared daily error:

$$IV_{i,m} = \sqrt{\frac{1}{T_{i,m}} \sum_{t=1}^{T_{i,m}} \varepsilon_{i,m,t}^2}, \quad (2)$$

where $T_{i,m}$ denotes the number of daily observations during a given month. The summary

statistics in Table 1 Panel A reveal mean (median) annualized total (TV) and idiosyncratic (IV) volatilities of 54.44% (42.02%) and 47.99% (36.36%) respectively.

At the beginning of every month, we sort all stocks by their idiosyncratic volatility during the previous month into five portfolios from low to high idiosyncratic volatility. We also construct a zero-investment portfolio (5-1) that invests in the highest IV portfolio by short-selling the lowest IV portfolio. For each portfolio p , we perform a time-series regression of monthly, value-weighted, excess returns on the monthly returns of the three Fama-French factors:

$$r_{p,m}^e = \alpha_p + \beta_p^{MKT} MKT_m + \beta_p^{SMB} SMB_m + \beta_{p,m}^{HML} HML_m + \varepsilon_{p,m} \quad (3)$$

Panel B of Table 1 reports for each portfolio the time-series mean and standard deviation of the total portfolio return as well as the average pricing error, α_p . The results are similar to the ones reported in Table VI in Ang et al. (2006): The Fama-French three factor model does not price high IV portfolios or the zero-investment arbitrage portfolio. In particular, the 5-1 portfolio produces an economically and statistically significant alpha of -1.16%.⁵ Ang et al. (2006, 2009) hypothesize that the pricing errors are due to a missing common factor, but their empirical investigations find that the errors are not explained by previously suggested additional pricing factors, such as momentum, skewness, liquidity, or market volatility.

2.2 Predicted and residual idiosyncratic volatility

To examine whether the observed pricing errors are possibly due to an omitted, but so far unexplored, systematic factor, we follow a new approach. We begin by decomposing a stock's idiosyncratic volatility into a component predicted by a small number of common factors that affect the idiosyncratic volatility of all stocks and a residual component that remains unexplained by these aggregate factors. We then repeat the analysis of Table 1 by sorting stocks into five portfolios based on a stock's predicted idiosyncratic volatility and five portfolios based on a stock's residual idiosyncratic volatility. The existence of commonality in idiosyncratic volatility does not necessarily imply the presence of omitted risk factors. Yet, if

⁵Ang et al. (2006) report a FF-3 alpha of -1.31% for portfolio 5-1. The difference between our and their results is most likely due to subsequent revisions of the Fama-French factors for the sample period.

the pricing errors reported in Table 1 are indeed due to one or more omitted factors, we expect to observe significant α_p 's for portfolios that are formed based on predicted idiosyncratic volatility, but not for portfolios that are formed on residual idiosyncratic volatility.

Formally, we hypothesize that the error term $\varepsilon_{i,m,t}$ in equation (1) contains a risk factor, $r_{m,t}^F$, such that $\varepsilon_{i,m,t} = \beta_{i,m}^F r_{m,t}^F + u_{i,m,t}$, where $u_{i,m,t}$ represents the truly idiosyncratic return component. The variance of $\varepsilon_{i,m,t}$ would therefore consists of exposure to a common component, $[\beta_{i,m}^F]^2 \text{Var}(r_{m,t}^F)$, and a residual component, $\text{Var}(u_{i,m,t})$. Sorting stocks based on the common component would separate stocks by their $[\beta_{i,m}^F]^2$. Both the magnitude and sign of $\beta_{i,m}^F$ are, however, important for our purposes. If an omitted factor accounts for the mispricing in Table 1, the sign of $\beta_{i,m}^F$ must be the same for most stocks in any portfolio with a significant alpha. Given that $r_{m,t}^F$ represents a systematic risk factor, it is likely that indeed most stocks with exposure to the factor have $\beta_{i,m}^F$'s of the same sign. The lowest contribution of $\text{Var}(r_{m,t}^F)$ to the variance of $\varepsilon_{i,m,t}$ would occur for stocks with $\beta_{i,m}^F$ close to zero. Indeed, consistent with little exposure to the missing risk factor, the evidence in Table 1 reveals that alpha, α_p , is close to zero for portfolios 1 through 3.⁶

2.2.1 Decomposing idiosyncratic volatility

We obtain common components for the cross-section of idiosyncratic volatility, by using the asymptotic principal component method of Connor and Korajczyk (1986, 1988).⁷

Assume that for the cross-section of N stocks over the time of M months the natural logarithm of idiosyncratic volatility is approximately generated by a linear model of k common components:

$$LIV = BF + \mu, \quad (4)$$

where LIV is a $N \times M$ matrix of log stock-level idiosyncratic volatilities, B is $N \times k$ matrix of loadings, F represents the $k \times M$ matrix of common components, and μ is the $N \times M$ matrix of unexplained variation in the logs of idiosyncratic volatilities. The common components F are obtained as the eigenvectors corresponding to the k largest eigenvalues of the matrix

⁶See MacKinlay (1995) and MacKinlay and Pastor (2000) for a related, but more formal, discussion.

⁷For a recent application to non-return data, see Korajczyk and Sadka (2008) who examine commonality in firm-level liquidity.

$\frac{LIV' LIV}{N}$. The time-invariant loadings B are found by (separately for each stock) regressing the log of the stock's monthly idiosyncratic volatility, $LIV_{i,m}$, onto the time-series of the k components, F_m :

$$LIV_{i,m} = B_i F_m + \mu_{i,m}. \quad (5)$$

These loadings allow us to decompose each stock's monthly idiosyncratic volatility into a predicted part, $\hat{IV}_{i,m} = \exp[\hat{B}_i F_m]$, and a residual part, $IV_{i,m}^{res} = IV_{i,m} - \hat{IV}_{i,m}$.⁸

Table 2 reports the amount of total, i.e. cross-sectional as well as time-series, variation in log-idiosyncratic volatility that is explained by variation in predicted log-idiosyncratic volatility, $\frac{\text{Var}(\hat{LIV})}{\text{Var}(LIV)}$, using up to five common factors. Interestingly, the first common component alone explains almost one third (32%) of the variation. Each additional common component increases the explained variation by only four to three percentage points.

Figures 1 through 3 plot the time series of the five common components. Because each of the components 2 through 5 appears to experience a structural break around 1984, the year of the introduction of NASDAQ listed securities into the CRSP database, we also plot the five common factors for a sample of NYSE stocks only.⁹

Figure 1 plots the time series of the first common component log idiosyncratic volatility for stocks listed on any exchange as well as only for those listed on the NYSE. The two time series are very similar. Indeed, the correlation between them is 0.895. The first common component for all exchanges is also highly correlated with value- and equal-weighted average idiosyncratic volatilities, -0.842 and -0.806, respectively, where the negative correlation reflects the fact that loadings (B) on the first components are negative and therefore imply a positive correlation between predicted and actual IV. The fact that the absolute value of the correlation with the value-weighted average is slightly higher is consistent with Ang et al.'s (2006) argument that the idiosyncratic volatility puzzle is not a small firm phenomena. Lastly, the first common component for stocks on all exchanges is also highly correlated with the value-weighted average of the total volatilities of all stock returns (-0.764) and the

⁸We work with logs of idiosyncratic volatilities to avoid the possibility of negative predicted IV in Regression 5.

⁹Investigating the factor structure of stock returns on NYSE and NASDAQ, Goyal, Prignon and Villa (2008) find that there are only two common pervasive factors that govern the returns for both exchanges, and there is at least one factor that is not the same across the two exchanges.

monthly volatility of daily returns on CRSP value-weighted market portfolio (-0.675).

Figure 2 presents the time-series of common log-idiosyncratic volatility components 2 through 5 for all stocks. Each of them appears to experience a structural break around 1984, the time of the introduction of NASDAQ listed securities, followed by a 10-year period of much higher volatility which subsides around 1994.¹⁰ Figure 3 plots the time-series of common components 2 through 5 for only NYSE stocks. None of them exhibits a similar structural break. A comparison of Figures 2 and 3 confirms that time-series of common components 2 through 5 for all three exchanges are dominated by the addition of NASDAQ listed securities to CRSP data.

2.2.2 Pricing of predicted and residual idiosyncratic volatility portfolios

Table 3 presents results for the pricing of five portfolios formed based on stocks' *predicted* idiosyncratic volatilities in the previous month, $\hat{IV}_{i,m-1}$, and a zero investment portfolio that is long the highest $\hat{IV}_{i,m-1}$ portfolio and short the lowest $\hat{IV}_{i,m-1}$ portfolio as well as the pricing of similar portfolios formed based on stocks' *residual* idiosyncratic volatilities, $IV_{i,m-1}^{res}$.

In Panel A of Table 3, we use only the first principal component to predict a stock's idiosyncratic volatility. The subpanel labeled "Predicted idiosyncratic volatility" reports the mean monthly excess return and its standard deviation as well as the alphas for each of the five portfolios as well as for the zero investment portfolio. The results are similar to the those reported in Panel B of Table 1 and Ang et al. (2006) for portfolios sorted on stocks' total idiosyncratic volatilities. That is, the mean excess returns first increases as we move from portfolio 1 to 3 and then decreases as we move from portfolio 3 to 5. The standard deviation, however, is strictly increasing, from 3.96% (portfolio 1) to 9.14% (portfolio 5). As before, we perform time-series regressions and report alpha, α_p , for each portfolio under the Fama-French three factor model. The results are again very similar to those in Table 1 Panel B. The alphas (and their t -statistics) for portfolios 1 through 4 are almost identical to the corresponding alphas of portfolios 1 to 4 in Table 1. Only Portfolio 5 has a noticeably

¹⁰Component 2 spikes at about 3,000 times its mean in November 1986. While component 3 takes on multiple extrema of similar absolute magnitude, factors 4 and 5 each reach unique extrema in September 1984 and December 1988, respectively.

different, and in fact, in absolute value larger alpha than the corresponding alpha in Table 1. The zero investment portfolio also yields a large negative alpha, which, again, is in absolute value larger than in Table 1. Hence, we find that portfolios formed only on the component of idiosyncratic volatility that is predicted by a stock's loading on the first principal component show very similar excess returns and alphas to those formed on total idiosyncratic volatility.

In the subpanel labeled “Residual idiosyncratic volatility,” we report corresponding results when forming monthly portfolios using the residual idiosyncratic volatility, $IV_{i,m-1}^{res}$, which accounts for about two thirds of total idiosyncratic volatility. The standard deviation of the lowest $IV_{i,m-1}^{res}$ portfolio is 5.11% relative to 6.22% for the highest $IV_{i,m-1}^{res}$ portfolio. The smaller spread in standard deviation between these two portfolios (relative to portfolios sorted on $\hat{IV}_{i,m-1}$) suggests that most of the variation of residual idiosyncratic volatility is stock-specific and, therefore, not apparent at the portfolio level. The alphas of portfolios 1 through 4 are indistinguishable from zero. Portfolio 5 and the zero investment portfolio yield alphas that are statistically significantly negative, but their absolute values and t -statistics are much smaller than those of corresponding portfolios formed on either total or predicted idiosyncratic volatility. In particular, the zero investment has an alpha of only -0.22 compared to -1.16 and -1.28 for the zero investment portfolios sorted on total idiosyncratic volatility and predicted idiosyncratic volatility (and t -statistic of only -2.37 compared to -7.49 and -6.79).

We repeat the analysis using all five common components to predict a stock's idiosyncratic volatility. The results, reported in Panel B of Table 3, are almost identical to those in Panel A. The only noticeable differences are that the alpha of the zero investment portfolio formed on predicted idiosyncratic volatility is in absolute value slightly larger and the alpha of the zero investment portfolio formed on exposure to residual idiosyncratic volatility is in absolute smaller and no longer statistically significant.

3 Idiosyncratic volatility, market volatility, and default spread

Above we have provided empirical evidence consistent with an omitted risk factor explanation of the IV puzzle of Ang et al. (2006, 2009). In this section, we shed some light on the nature

of the omitted systematic risk. Our investigation builds upon the study of Ferson and Harvey (1999) which advances that the Fama-French 3-factor model does not fully capture the effects of business cycles on expected stock returns.

Table 4 presents results relating the first common component of log IV to several economic state variables, using monthly data between August 1963 and December 2000. Specifically, in column 1, we report results from a regression of changes in the first common component onto a constant and changes in standard macro-economic state variables: the term as well as the default spread,¹¹ the dividend yield on the S&P 500 index, and the monthly volatility of daily CRSP market returns. We also include (in levels as opposed to first differences) the NBER recession indicator, which equals one during recessions and zero otherwise. Given the negative correlation between the first common component and average idiosyncratic volatility and the negative loadings (B), we have changed the sign of first component for ease of interpretation. Changes in the first common component are then significantly positively associated with changes in the default spread as well as changes in the market volatility. The resulting adjusted R^2 is 0.30. Hence, the regression suggests that idiosyncratic volatility, calculated using the Fama-French three factor model, contains a common component whose time-series variation can be partially explained by standard business cycle proxies, specifically the default spread. This is consistent with the view that the common component of idiosyncratic volatility represents systematic risk, which is omitted by the Fama-French model, and that stocks with large exposure to this systematic risk reflect such exposure in their expected returns.

Chen and Petkova (2012) observe that the stock market variance can approximately be expressed as the product of two components: the average variance of all the stocks and the average correlation among all the stocks. They find that in the cross-section of IV portfolios changes in the average stock return variance are priced negatively, but changes in the average stock return correlation are not. They attribute the negative premium of high IV stocks to the fact that due to their growth options they serve as a hedge against

¹¹The term spread is yield difference between 10-year and 1-year U.S. Treasuries, while the default spread captures the yield difference between BAA and AAA rated U.S. corporate bonds. Both variables are among the asset market indicators used to predict real economic indicators (see, e.g., Harvey (1988) and Gertler and Lown (1999)) and have been used as proxies for risk factors in asset pricing tests (for example, Chen, Roll and Ross (1986), Fama and French (1993), and Hahn and Lee (2006)).

increases in the average equity return variance. We therefore replace aggregate market volatility with average stock volatility and average stock correlation (measured as aggregate market volatility divided by average stock volatility). Column 2 reports the results. The adjusted R^2 increases from 0.30 to 0.35, and changes in both the average stock volatility and the average stock correlation as well as changes in the default spread are significant determinants of changes in the first common IV component. Economically, a one standard deviation of in the average variance (0.1656) is associated with a 0.49 standard deviation of changes in the first common IV component, while a one standard deviation of default spread changes (0.1141) lead to 0.10 standard deviations of changes in the common component. This result shows an economic dimension by which the first common component of log IV is different from the average stock volatility and average stock correlation. Specifically, the first component of IV captures variations in the default spread that is orthogonal to variation in average volatility.

In column 3, we replace the first differences of average stock volatility and the average stock correlation with the corresponding innovations that Chen et al. (1986) obtain from a vector autoregressive (VAR) approach. This specification, which is estimated over a slightly shorter sample from July 1996 to December 2000, reveals that changes in the first common component are significantly related to changes in the default spread and the dividend yield as well as innovations in average variance. The common component therefore seems to be related to both, the investment opportunity set and the business cycle, in particular as captured by the default spread. In fact, the relationship between the first component of log IV and the default spread is statistically significant in all three specifications.

4 The predicted idiosyncratic volatility factor (PIV)

So far, we have shown that the part of idiosyncratic volatility that is related to an economically and statistically significant alpha reflects exposure to common factors of idiosyncratic volatility. This, by itself, does not mean that common factors of idiosyncratic volatility are systematic risk factors. If a common factor of idiosyncratic volatility is priced because it reflects exposure to one or more missing systematic risk factors, a factor based on predicted

idiosyncratic volatility should contribute to explaining the cross-section of stock returns. In this section, we study this question. In particular, we analyze whether the return to a zero investment portfolio that is long stocks with high predicted idiosyncratic volatility and short stocks with low predicted idiosyncratic volatility plays a role in pricing excess returns on the 30 Fama-French industry portfolios. Using industry portfolios instead of the Fama-French 25 size-B/M portfolios reduces the risk of spurious results due to the strong factor structure of size-B/M portfolios (Lewellen, Nagel and Shanken (2010)).¹²

To avoid any look-ahead bias, we construct our predicted idiosyncratic volatility factor out-of-sample, i.e. using only past data. As before, we perform our analysis using one as well as five common components of log IV. We refer to the resulting predicted idiosyncratic volatility factors as $PIV(1)$ and $PIV(5)$. Specifically, to create say, $PIV(5)$, we start by using the first 67 observations of monthly idiosyncratic volatility (from July 1963 until January 1969) based on the residuals from equation (1). We extract five monthly log IV components and estimate each stock i 's loading onto these components for that period, B_i^{67} , where the superscript reflects the use of the first 67 months of data. A stock's predicted idiosyncratic volatility for January 1969 is then given by $\exp[B_i^{67} F_{196901}]$. Sorting on the stocks predicted idiosyncratic volatilities in January 1969, we form a zero investment (highest minus lowest quintile) portfolio and calculate its value-weighted return during February 1969, which is the first observation of $PIV(5)$. We repeat this process until we obtain idiosyncratic volatility predictions from January 1969 until November 2000, for each month using all data available between July 1963 and the end of that month. Using the out-of-sample predicted idiosyncratic volatilities, we form monthly zero investment portfolios and calculate their returns in the subsequent month, from February 1969 until December 2000.

In order to evaluate the contribution of PIV in pricing the cross-section of risky assets, we begin by evaluating the performance of a linear pricing kernel, consisting of the three Fama-French factors and $PIV(1)$ or $PIV(5)$:

$$M_m = 1 + b^{MKT} MKT_m + b^{SMB} SMB_m + b^{HML} HML_m + b^{PIV} PIV_m. \quad (6)$$

¹²Nevertheless, we also repeated the tests in this section using the Fama-French 25 size-B/M portfolios. Our conclusions remain the same.

Using monthly excess returns on the 30 industry portfolios, we estimate the parameters of the pricing kernel and evaluate the model performance using iterative GMM estimation and exploiting the standard (unconditional) asset pricing equation:

$$E[M_m r_{p,m}^e] = 0. \quad (7)$$

Table 5 reports estimates of the model parameters, b^{MKT} , b^{SMB} , b^{HML} , and b^{PIV} , the price of risk corresponding to each factor as well as statistics related to model fit. In particular, we use Hansen's (1982) J -statistic and the Hansen and Jagannathan (1997) distance, HJ , to assess overall model fit. In column 1, we report the Fama-French three factor model as a baseline case. All three factors significantly contribute to the pricing of the 30 industry portfolios as the estimates of the pricing kernel coefficients are statistically significant. All three are also priced, with MKT carrying a significantly positive risk premium, while HML and SMB have a significantly negative price of risk during our sample period. Next, we add $PIV(1)$ to the pricing kernel. As explained above, $PIV(1)$ is the return on the portfolio of stocks whose predicted (by the first common IV factor) idiosyncratic volatility is high minus the return on the portfolio of stocks whose predicted idiosyncratic volatility is low. Column 2 of Table 5 reports the results. The estimate of the coefficient on $PIV(1)$ is significantly different from zero, suggesting that $PIV(1)$ helps to price our test assets and should be included in the pricing kernel. The inclusion of $PIV(1)$ drives out the SMB factor, while MKT and HML remain significant. The magnitude of the risk premium on $PIV(1)$ of -111 basis points (bp) per month is similar to the mean return of the 5-1 portfolio formed on predicted idiosyncratic volatility in Table 3 Panel A. The SMB risk premium is -47 bp and the HML risk premium -22 bp, while the market risk premium is 73 bp. Finally, the Hansen's J -statistic confirms that adding $PIV(1)$ improves the fit of the pricing kernel, as the J -statistic drops from 150 to 125. The Hansen-Jagannathan distance, on the other hand, only drops slightly. Both statistics still reject the augmented pricing kernel, based on the size of the pricing errors. We repeat the test using $PIV(5)$ instead of $PIV(1)$. The results, reported in the third column, are very similar.

Although we find that $PIV(1)$ and $PIV(5)$ represent one or more return factors omitted

by the Fama-French three factor model, there is existing empirical evidence that there are two other factors that may be missing from the Fama-French model: momentum and liquidity (see Carhart (1997), Pastor and Stambaugh (2003), Sadka (2006) and Korajczyk and Sadka (2008)). We, therefore, examine whether $PIV(1)$ and $PIV(5)$ are simply proxies for these effects. We add the monthly risk premiums on momentum (MOM) and Pastor-Stambaugh traded liquidity factor (LIQ) to the pricing kernel in equation (6). The results in columns 4 and 5 of Table 5 show a significant coefficient for MOM , but an significant coefficient for LIQ . But, importantly, MOM and LIQ do not drive out either $PIV(1)$ or $PIV(5)$, as their coefficients remain significant. The GMM tests, which use industry portfolios, i.e., portfolios that are not sorted by IV, as test assets, thus suggest that PIV is a priced risk factor and not a firm characteristic.

5 PIV and return anomalies

Is it possible that other asset pricing anomalies can be explained by the same common factors we uncovered to explain the IV puzzle? In this section, we address this question for five return anomalies that earn abnormal (excess) returns with respect to the five factor model above, which includes the Fama and French (1993) three factors, momentum and the Pastor-Stambaugh traded liquidity factor. In particular, we examine the returns of zero investment trading strategies based on earnings surprises (see, for example, Sadka (2006)), financial distress (see, for example, Campbell et al. (2008)), net stock issues (see, for example, Fama and French (2008)), profitability (see, for example, Fama and French (2008)), and illiquidity (see, for example, Cochrane (2011)).¹³

¹³The earnings surprise anomaly (also known as the post-earnings announcement drift anomaly) describes the puzzle, documented by Ball and Brown (1968), Foster, Olsen and Shevlin (1984), and more recently Sadka (2006), of an abnormal positive difference between the average returns of firms with high unexpected earnings and firms with low unexpected earnings. The financial distress anomaly describes the negative relation between credit risk and return, documented by Dichev (1998) and Campbell et al. (2008), which is manifested in an abnormally positive average return on a portfolio of least minus most financially distressed firms. The net stock issues anomaly, documented by Loughran and Ritter (1995) and most recently studied by Fama and French (2008), describes the finding that equity issuing firms underperform matching non-issuing firms in the post-issue period. The profitability anomaly, studied most recently by Fama and French (2008), describes the puzzle that highly profitable firms generate abnormally higher average returns than highly unprofitable firms. The illiquidity anomaly describes the puzzle that stocks with higher illiquidity levels earn abnormally higher average returns than stocks with lower illiquidity levels, even after controlling

Following Fama and French (2008), we examine both value-weighted and equal-weighted returns. For each anomaly, Table 6 reports the sample period as well as the average monthly returns and t -statistics. The average monthly (excess) returns and their t -statistics are larger (in absolute value) for equal-weighted than value-weighted returns. Aside from the value-weighted financial distress returns, all the average returns are significantly different from zero. The profitability returns are from Chen, Novy-Marx and Zhang (2011) and only available as value-weighted returns. For the illiquidity anomaly, we report equal-weighted returns only, because the corresponding value-weighted portfolio does not exhibit significant mispricing over our sample period.

5.1 Correlations between return anomalies

To begin our investigation, we run time series regressions of the value- and equally-weighted anomaly returns onto the five factor model consisting out of the three Fama French factors as well as the momentum and liquidity factor. If exposure to systematic risk is unaccounted for, the residuals of these regressions, i.e. the unexplained return component, should be correlated across the different anomalies. Table 7 reports coefficients of pairwise correlations between residual returns for all anomalies, using value-weighted returns (Panel A) as well as equally-weighted correlations (Panel B). In particular, the entries above the diagonal reflect residual from the five factor model, while entries below the diagonal reflect correlations between residuals from regressions in which the five factor model has been augmented by including $PIV(1)$, our empirical proxy for the unaccounted common factor behind the IV puzzle. For simplicity, we henceforth refer to $PIV(1)$ as PIV .¹⁴ Without including PIV in the pricing model, the correlations are large and often highly statistically significant (p -values are reported below each correlations coefficient). Indeed, the average absolute correlation of the residuals of the four value-weighted return anomalies is 30%, while it is 36% for the four equal-weighted return anomalies. Controlling for PIV reduces the absolute correlation

for many types of liquidity factors (see, for example, Acharya and Pedersen (2005)).

¹⁴Because the results for $PIV(1)$ and $PIV(5)$ are very similar, to keep the discussion of our results as focused as possible, we only report results for $PIV(1)$, i.e. the zero investment portfolio formed by using idiosyncratic volatility predicted by only the first common IV component. The results for $PIV(5)$ are available upon request.

of pricing errors by about one half to 14% for the value-weighted return anomalies, and by about one third to 24% for the equal-weighted return anomalies. We conclude that time series variation of the residual returns of several existing return anomalies exhibit substantial commonality that, to a substantial part, is related to our risk proxy *PIV*.

5.2 *PIV* as a missing risk factor

Table 8 reports the details of the time series regressions that we run for each anomaly. We report the coefficient estimates for all factors, the amount of average mispricing (alpha) in terms of the constant of the time series regressions, and the adjusted R^2 of each regression. Panel A reports results when using value-weighted returns of the anomaly. The first two columns present the results for the post-earnings-announcement drift anomaly, the third and fourth columns present the results for the financial distress anomaly, the fifth and six columns present the results for the net stock issues anomaly, and the last two columns present the results for the profitability anomaly. The first column in each pair presents the results for the five-factor model without *PIV*. The constant is significantly different from zero in each case. The second column in each pair shows that adding *PIV* has a significant effect. The coefficient on *PIV* is always significantly different from zero (p -value < 0.00001). In each case, adding *PIV* substantially reduces the absolute value of alpha and significantly improves the empirical fit. In the case of the post-earnings-announcement anomaly, adding *PIV* reduces alpha by one-third, from 30 to 20 basis points (bp) per month, and alpha is no longer statistically significant at conventional levels. In addition, the adjusted R^2 increases from 0.06 to 0.09. The impact of *PIV* is especially noticeable in the case of the financial distress anomaly reducing its alpha falls by 77% from 94 to 22 bp per month, which is no longer statistically significantly different from zero. Furthermore, the adjusted R^2 more than triples, from 0.12 to 0.44. In the case of the net issues anomaly, the adjusted R^2 increases from 0.22 to 0.35, and the absolute value of alpha falls by 25% from 80 to 60 bp per month but remains statistically significant. Finally, in the case of the profitability anomaly, the adjusted R^2 increases by more than 5 times, from 0.06 to 0.41, and alpha falls by one-third from 91 to 60 bp per month, but remains statistically significant.

Panel B reports the results for equal-weighted returns. The first six columns present the results for the anomalies that are reported in the corresponding value-weighted columns and the last two columns present the results for the illiquidity anomaly. Similar to the value-weighted anomalies, we find that adding *PIV* to the pricing kernel helps explain the anomalies. As the second column in each pair shows, the coefficient on *PIV* is always significantly different from zero ($p\text{-value} < 0.00001$) and adding *PIV* results in a higher adjusted R^2 . In the case of the post-earnings-announcement anomaly, adding *PIV* has a relatively small effect: the monthly alpha falls from 142 to 135 bp only and it remains statistically significant at conventional levels. In addition, the adjusted R^2 increases from 0.15 to 0.18. In the case of the financial distress anomaly, alpha again falls substantially (by 62%) from 111 to 42 bp per month, though it remains statistically significant; and the adjusted R^2 almost quadruples, from 0.16 to 0.62. In the case of the net issues anomaly, the absolute value of alpha falls by about one-third, from 98 to 68 bp per month, though it also remains statistically significant. The adjusted R^2 doubles from 0.25 to 0.52. Finally, the impact of *PIV* is also noticeable in the case of the illiquidity anomaly, in which alpha falls by more than one-half, from 56 to 27 bp per month and becomes statistically insignificant at conventional levels. The adjusted R^2 triples from 0.12 to 0.37.

Hence, we find clear evidence that adding *PIV* to an asset pricing model that includes the three Fama-French factors, momentum, and the Pastor-Stambaugh traded liquidity factor, significantly improves the ability to explain each of the puzzles (e.g., the adjusted R^2 increases by about 23 percentage points on average) and reduces the correlation of pricing errors between the puzzles. The abnormal returns associated with zero investment portfolios based on sorts of earnings surprises, financial distress, net stock issues, profitability and illiquidity decline substantially, on average by about 40%. In the cases of value-weighted financial distress and earnings surprises, and equal-weighted illiquidity, the intercepts are no-longer significantly different from zero. Among all these anomalies, the economic effect of controlling for *PIV* is mostly noticeable in the financial distress anomaly. This finding reinforces the relationship between default spread and *PIV* documented in Table 4.

In unreported results, we also compare the performance of *PIV* to using the 5-1 IV

portfolio return (as constructed by Ang et al. (2006)) as a factor.¹⁵ The results reveal that *PIV* dominates the 5-1 IV portfolio return. In all the cases, when both of them are included in the above time series regressions, the coefficient on the 5-1 IV portfolio return is never significantly different from zero. These findings suggest that sorting stocks based on their exposure to the common component of IV factor yields a better proxy for the missing systematic risk than sorting stocks on total IV, which includes the volatilities of both the missing systematic risk and the remaining idiosyncratic risk.

6 Robustness

We report three sets of robustness checks. First, we examine the role of microstructure effects on idiosyncratic volatility following Han and Lesmond (2011). Second, we confirm that *PIV* is priced when the test assets exclude low credit quality firms (Avramov et al. (2013)). Finally, we document that *PIV* matters for asset pricing even in the presence of factors that mimic innovations in average variance and average correlation (Chen and Petkova (2012)).

6.1 Measuring idiosyncratic volatility

Han and Lesmond (2011) show that controlling for microstructure effects, such as bid-ask spreads, in the estimation of idiosyncratic volatility reduces and in some cases eliminates the ability of idiosyncratic volatility to predict future returns. In particular, Han and Lesmond (2011) estimate idiosyncratic volatility using quote-midpoint prices as opposed to close-to-close prices as in Ang et al. (2006). Using data between 1985 and 2000, Panel A of Table 9 replicates their finding of a reduced (absolute) alpha when portfolios are formed on idiosyncratic volatility estimated from quote-midpoint prices.¹⁶ We next perform the same decomposition of this alternative measure of idiosyncratic volatility as above. In Panel B of Table 9, we report results for portfolios based on predicted as well as residual idiosyncratic volatility. Importantly, the alpha of going long the high predicted IV portfolio and short

¹⁵The two factors are highly correlated with one another (about 89.0%).

¹⁶All results in Table 9 are based on IV estimates using quote-midpoint prices from Han and Lesmond (2011). We thank David Lesmond for making the data available to us.

the low predicted IV portfolio increases in absolute value, going from - 57 bp in Panel A to -93 bp. At the same time, the corresponding strategy applied to the residual IV portfolios now yields a significantly positive alpha of 54 bp. While micro-structure effects seem to somewhat confound the IV results based on close-to-close prices, using quote-midpoint prices to measure IV and sorting stocks based on their exposure to common IV components recovers a significantly negative and sizable premium. Overall, we interpret these results as consistent with an omitted risk factor.

In addition, we also compare the ability of *PIV*, which we estimated using close-to-close prices, and the corresponding zero-investment (high-minus-low) predicted IV portfolio, which we estimated using quote-midpoint prices, to explain the returns anomalies studied in Table 8. We report the results in Table 10 for Feb. 1985-Dec. 2000, which is the sample period for which both returns series are available. The two *PIV* returns series yield similar results when added to the 5-factor model of the Fama and French (1993) three factors, momentum and the Pastor-Stambaugh traded liquidity factor. The coefficients of either series are always significant at conventional levels and their magnitudes are similar. The magnitude and statistical significance of the abnormal returns are also similar, although the mid-quote *PIV* typically yields lower estimated intercepts. Lastly, the mid-quote *PIV* typically yield lower adjusted R-squared values.¹⁷ Hence, it does not appear that our findings are driven by using close-to-close prices rather than quote-midpoint prices.

6.2 Low credit quality firms

Avramov et al. (2013) suggest that the IV puzzle and other asset pricing anomalies occur mostly in low credit quality firms.¹⁸ In particular, they argue that when firms enter into financial distress, idiosyncratic volatility increases and, at same time, negative stock returns are realized due to the expected costs of distress and credit downgrades. They conclude that

¹⁷One difference between the two *PIV* series in Table 10 is that while the close-to-close *PIV* series is out-of-sample, the mid-quote *PIV* series is within sample. Because the sample period that matches the Han and Lesmond (2011) sample is less than one-half of the sample period studied in Table 8, we use within-sample mid-quote *PIV* returns to avoid losing additional 5 years of data.

¹⁸Note that Avramov et al. (2013) follow Campbell et al. (2001) and calculate idiosyncratic returns as the stock's return less the market return. We follow Ang et al. (2006) and allow for stock-specific exposure to all three Fama French factors.

the IV puzzle therefore reflects a firm characteristic, rather than exposure to a systematic factor. According to this result, we expect that *PIV* matters for asset pricing tests if the test assets include low credit quality firms, but becomes insignificant and does not command a (negative) risk premium when low credit quality stocks are excluded from the test assets.

We therefore repeat our main GMM asset pricing test from Table 5, but exclude firms with low credit quality. In particular, we form 30 Fama French industry portfolios excluding firms in the lowest credit rating tercile (as determined by Avramov et al. (2013)). In Table 11, we first report results when , as above but for a shorter time period, using the 30 Fama French industry portfolios (column 1). In column 2, we report results after low credit rating firms have been dropped from the industry portfolios.

We find that *PIV* continues to play a significant role in the pricing of the industry portfolios and, as before, commands a negative risk premium. That is, even when excluding those firms which according to Avramov et al. (2013) are the source of the IV puzzle from the test assets, we find that *PIV* contains relevant information for the pricing of the remaining assets, consisting of all firms that have a higher credit rating or no credit rating at all. In untabulated results, we repeat the estimation including only firms with a credit rating, but again excluding those in the lowest credit rating tercile. As in column 2, we find that *PIV* is a significant part of the pricing kernel and has a significantly negative risk premium.

6.3 Controlling for innovations in average variance and in average correlation

We compare the performance of our empirically motivated risk factor, *PIV*, to innovations of the average variance and correlation of Chen and Petkova (2012). Chen and Petkova (2012) show that high IV firms have high research and development expenditures and they attribute the negative premium of high IV stocks to the fact that due to their growth options they serve as a hedge against increases in the average equity return variance.

Column 3 of Table 11 reports our main result from Table 5 for comparison. The pricing kernel includes the three Fama-French factors, momentum, liquidity, and *PIV*. Column 4 adds *PAV* and *PAC*, the monthly excess returns on the mimicking portfolios for innovations

in average variance and average correlation from Chen and Petkova (2012). The correlation between PIV and PAV and between PIV and PAC are -34% and 19%, respectively. These results suggest that PIV is not a proxy for PAV and PAC . The coefficient on PIV remains statistically significant when adding PAV and PAC to the pricing kernel. The premium on PIV , however, drops by about 20%, but remains significantly different from zero. The data suggest, however, that both PAV and PAC belong in the pricing kernel as well. Their coefficients are significantly negative and their inclusion leads to clear reductions in both the J - and HJ -statistics, with p -values of 37% and 10% respectively. Lastly, as in Chen and Petkova (2012) who use returns on 49 industry portfolios as test assets, the premium on PAV is not statistically significant, while opposite to Chen and Petkova (2012) the premium on PAC is significantly positive.

Finally, we examine to what extent adding PAV and PAC helps resolve the return anomalies studies in Section 5. We report these results in Table 12. The first column of each anomaly reports results from Table 8 for regressions in which PIV is added to the five factors model. The second column of each anomaly reports the regression which also includes PAV and PAC in addition to PIV .

The results suggest that adding PAV and PAC does not drive out PIV as a significant factor in explaining the return anomalies. Though adding PAV and PAC typically reduces the estimated coefficient on PIV , PIV remains statistically significant at conventional levels in each case. That is, all return anomalies are significantly related to variation in PIV that is orthogonal to variation in PAV and PAC .

Moreover, PAV and PAC are often statistically insignificant and do not reduce the abnormal return for most anomalies. For the value-weighted anomalous returns, the coefficients on PAV or PAC are statistically significant only for the net stock issues anomaly, when both are significantly negative. On the other hand, for the equally-weighted return anomalies, the coefficients on PAC are never statistically significant, whereas the coefficients on PAV are significantly negative for the financial distress and illiquidity anomalies, and positive (though with a t -statistic of 1.81) for the net stock issues anomaly.

7 Conclusions

We examine the hypothesis that systematic risk, which is unaccounted for by the Fama-French model, explains the significant difference in returns of stocks with high idiosyncratic volatility and stocks with low idiosyncratic volatility. We show that the cross-section of monthly (Fama-French based) idiosyncratic return volatility commoves substantially. A principal component analysis reveals that exposure to a single common component of (log) idiosyncratic volatility explains 32% of the variation and exposure to five common components explains almost 50% of the variation. Finally, we find that time-series variation in the first common IV component is related to business cycle proxies, in particular to default spread and market volatility. When forming portfolios based on the idiosyncratic volatility predicted by the common IV component(s), we find that almost all of the return differential documented by Ang et al. (2006) occurs in these portfolios. Investment portfolios formed on stocks' exposures to residual idiosyncratic volatility no longer exhibit significant abnormal returns.

Using the return difference between a portfolio of high predicted idiosyncratic volatility and a portfolio of low predicted idiosyncratic volatility, we construct a new risk factor (*PIV*) and show that, in the presence of the three Fama-French factors, *PIV* significantly contributes to explaining the cross-section of returns. Formal tests reveal that *PIV* should be added to existing linear factor models. We also find that *PIV* is different from momentum and liquidity, which the existing literature suggests are possible factors that are missing from the Fama-French model. Furthermore, we find that adding *PIV* to an asset pricing model that includes the three Fama-French factors, momentum and liquidity, helps explain the earnings surprise, financial distress, net stock issues, profitability, and illiquidity anomalies. Among all these anomalies, the economic effect of controlling for *PIV* is particularly important in the financial distress anomaly. This finding reinforces the finding that *PIV* captures information in the default spread.

We show that our findings are robust to using idiosyncratic volatility estimated from quote-midpoint prices as proposed by Han and Lesmond (2011), to controlling for factors that mimic innovations in average variance and in average correlation as in Chen and Petkova

(2012) and to excluding low credit quality firms (e.g., Avramov et al. (2013)).

Overall, our results reconcile the apparent contradictory interpretations of the IV puzzle by Chen and Petkova (2012) and by Avramov et al. (2013). Chen and Petkova (2012) attribute the low return of high IV stocks to the fact that they serve as a hedge to increases in market variance. Avramov et al. (2013), on the other hand, attribute the IV puzzle to a firm characteristic related to low credit quality. Our results point out that the IV puzzle is due to a missing risk factor, not a firm characteristic. But we show that the missing risk factor is related to the average stock volatility as well as the defaults spread and hence financial distress.

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Table 1

Idiosyncratic volatility

July 1963 - December 2000

Panel A: Summary statistics

Annually	Mean	Median	Std Dev
TV	0.5444	0.4202	0.4790
IV	0.4799	0.3636	0.4390

Panel B: Replication of Ang et al. (2006)

Rank	Mean	Std Dev	FF-3 Alpha
1	1.06	3.90	0.06 (1.29)
2	1.17	4.80	0.08 (1.24)
3	1.24	5.69	0.07 (0.82)
4	0.94	6.84	-0.34 (-3.53)
5	0.34	7.86	-1.10 (-8.31)
5-1	-0.72		-1.16 (-7.49)

Table 1 Panel A presents the mean, median, and standard deviation of firm-level annualized total (TV) as well as idiosyncratic (IV) return volatility. Panel B replicates results reported in Table VI of Ang et al. (2006). Every month, stocks are sorted into five portfolios based on the stock's previous month idiosyncratic volatility. For each portfolio (1 through 5) we report (in percentage terms) the mean monthly return, standard deviation, as well as alpha with respect to the Fama–French (1993) three factor model. 5-1 refers to a strategy that is long the portfolio of stocks with the highest idiosyncratic volatility in the previous month and short the portfolio of stocks with the lowest idiosyncratic volatility in the previous month. For each estimate of alpha, we also report the corresponding robust (Newey–West with four lags) *t*-statistic.

Table 2
Commonality in idiosyncratic volatility
 July 1963 - December 2000

Number of Components	1	2	3	4	5
$\frac{Var(\hat{IV})}{Var(IV)}$	32.00%	36.49%	39.92%	42.97%	45.61%

Table 2 reports the amount of total, i.e. cross-sectional as well as time-series, variation in firm-level idiosyncratic return volatility that is explained by variation in predicted idiosyncratic return volatility, using between one and five common components of idiosyncratic return volatility.

Table 3

Return statistics of portfolios formed on predicted and residual idiosyncratic volatility

July 1963 - December 2000

Panel A: One common component

Predicted idiosyncratic volatility				Residual idiosyncratic volatility			
Rank	Mean	Std Dev	FF-3 Alpha	Rank	Mean	Std Dev	FF-3 Alpha
1	1.05	3.96	0.05 (1.25)	1	0.93	5.11	0.06 (1.53)
2	1.14	5.29	0.01 (0.19)	2	1.01	5.57	0.06 (1.55)
3	1.22	6.55	0.06 (0.62)	3	0.99	5.58	0.01 (0.45)
4	0.93	7.91	-0.34 (-2.48)	4	1.05	6.01	0.00 (0.03)
5	0.07	9.14	-1.24 (-7.42)	5	0.93	6.22	-0.16 (-2.35)
5-1'	-0.98		-1.28 (-6.79)	5-1'	-0.01		-0.22 (-2.37)

Panel B: Five common components

Predicted idiosyncratic volatility				Residual idiosyncratic volatility			
Rank	Mean	Std Dev	FF-3 Alpha	Rank	Mean	Std Dev	FF-3 Alpha
1	1.06	3.95	0.06 (1.53)	1	0.96	4.35	-0.01 (-0.17)
2	1.15	5.23	0.01 (0.27)	2	1.21	4.49	0.19 (3.08)
3	1.19	6.53	-0.01 (-0.11)	3	1.13	4.47	0.09 (1.89)
4	0.88	8.00	-0.42 (-3.14)	4	1.08	4.67	-0.03 (-0.58)
5	0.03	9.19	-1.34 (-7.36)	5	1.05	5.05	-0.15 (-1.72)
5-1'	-1.03		-1.39 (-6.80)	5-1'	0.09		-0.14 (-1.17)

Table 3 presents return statistics of portfolios formed on predicted and residual idiosyncratic volatility. Every month, stocks are sorted into five portfolios based on the stock's predicted or residual idiosyncratic volatility during the previous month. For each portfolio (1 through 5) we report (in percentage terms) the mean monthly return, standard deviation, as well as alpha with respect to the Fama–French (1993) three factor model. 5-1 refers to a strategy that is long the portfolio of stocks with the highest idiosyncratic volatility in the previous month and short the portfolio of stocks with the lowest idiosyncratic volatility in the previous month. For each estimate of alpha, we also report the corresponding robust (Newey–West with four lags) *t*-statistic. Panel A reports results when only one common component is used to predict idiosyncratic volatility. Panel B reports results when five common components are used to predict idiosyncratic volatility.

Table 4

Economic content of common (log) idiosyncratic volatility

	1	2	3
Constant	0.0024 (0.0000)	0.0014 (0.0000)	-0.0013 (-0.2293)
First Difference of Term Spread	0.0033 (0.2909)	0.0066 (0.8023)	0.0020 (0.2263)
First Difference of Default Spread	0.0981 (2.4752)	0.0877 (2.3655)	0.0717 (2.5483)
First Difference of Dividend Yield	0.0149 (0.3615)	0.0114 (0.2985)	0.0807 (1.9831)
NBER Recession Indicator	-0.0029 (-0.4161)	-0.0011 (-0.1730)	-0.0038 (-0.4222)
First Difference of Market Volatility	0.1507 (12.8257)		
First Difference of Average (Total) Stock Volatility		0.3180 (4.9646)	
First Difference of Average Stock Correlation		0.0958 (4.2207)	
Innovations in Average Variance (Chen and Petkova)			1.5344 (7.0050)
Innovations in Average Correlation (Chen and Petkova)			0.1954 (1.0979)
Adj. R^2	0.3033	0.3486	0.3607
Sample Period	Aug. 1963 - Dec. 2000	Aug. 1963 - Dec. 2000	July 1966 - Dec. 2000

Table 4 reports results for linear regressions of the first difference of the first common component of log IV (multiplied by minus one) onto the first difference of several economic state variables (Term Spread, Default Spread, Dividend Yield, Market Volatility, Average (Total) Stock Volatility, Average Stock Correlation, the NBER Recession Indicator (1 during recessions and zero otherwise), and innovations in Average Variance and Average Correlation. For each variable, we report the coefficient estimate and robust (Newey-West with up to 12 lags) t-statistics.

Table 5
GMM results using 30 industry portfolios
February 1969 - December 2000

		1	2	3	4	5
C O E F F I C I E N T	<i>MKT</i>	-2.7198 (-8.3115)	-5.1504 (-9.8876)	-4.2099 (-10.0852)	-5.3312 (-6.2338)	-5.0836 (-6.2883)
	<i>SMB</i>	5.0924 (9.7368)	-1.3219 (-0.9839)	-1.0657 (-0.8352)	-0.8200 (-0.3033)	-0.9209 (-0.3206)
	<i>HML</i>	3.4277 (6.4048)	2.8831 (3.6130)	2.5035 (3.6785)	-5.8182 (-3.2192)	-5.4671 (-3.0485)
	<i>PIV (1)</i>		4.2376 (5.9667)		2.6740 (2.1172)	
	<i>PIV (5)</i>			3.4530 (5.3765)		2.5901 (1.9886)
	<i>MOM</i>				-12.7440 (-6.1335)	-12.2415 (-5.9670)
	<i>LIQ</i>				-2.0693 (-1.0408)	-2.5429 (-1.3127)
P R E M I U M	<i>MKT</i>	0.0056 (8.9260)	0.0073 (8.0304)	0.0063 (8.3709)	0.0064 (3.4417)	0.0063 (3.5551)
	<i>SMB</i>	-0.0034 (-7.2628)	-0.0047 (-6.7644)	-0.0043 (-6.9766)	-0.0048 (-2.7124)	-0.0050 (-2.9225)
	<i>HML</i>	-0.0032 (-7.3034)	-0.0022 (-3.3757)	-0.0021 (-3.8479)	0.0005 (0.3959)	0.0003 (0.2222)
	<i>PIV (1)</i>		-0.0111 (-6.8027)		-0.0087 (-2.3015)	
	<i>PIV (5)</i>			-0.0115 (-7.0356)		-0.0119 (-2.9154)
	<i>MOM</i>				0.0190 (6.0447)	0.0185 (5.9802)
	<i>LIQ</i>				0.0021 (0.8201)	0.0027 (1.0590)
	<i>J</i>	149.9744 [0.0000]	124.8686 [0.0000]	132.5829 [0.0000]	37.9532 [0.0351]	39.1285 [0.0265]
	<i>HJ</i>	0.2250 [0.0000]	0.2238 [0.0000]	0.2231 [0.0000]	0.1829 [0.0018]	0.1860 [0.0006]

Table 5 presents results from iterative GMM estimation of different asset pricing models. The tests assets are monthly excess returns of the 30 industry portfolios. The sub panels "Coefficient" and "Premium" list the possible factors (market (*MKT*), size (*SMB*), value (*HML*), predicted *IV* (*PIV*), momentum (*MOM*), and liquidity (*LIQ*)) considered in a linear pricing kernel. *PIV (1)* refers to the case in which *IV* is predicted using the first common *IV* component, whereas *PIV (5)* refers to the case in which the first five common components are used. Each column considers a different linear combination of these factors. We report the coefficient estimate for each factor in the pricing kernel as well as the corresponding risk premium associated with the factor. Robust (Newey-West (1987) with one lag) *t*-statistics are in parentheses below the estimated coefficients and premiums. We report Hansen's (1982) *J*-statistic and Hansen-Jagannathan (1997) distance (*HJ*). *p*-values are in brackets below these test-statistics.

Table 6

Description of return anomalies

Anomaly	Description	Sample period	Average Monthly Return	
			Value-weighted	Equally-weighted
Earnings Surprises	Returns of highest minus lowest decile portfolios of all NYSE, AMEX and NASDAQ stocks in the CSRP-Compustat intersection, sorted by standardized unexpected earnings (see Sadka (2006)).	Jan. 1972 - Dec. 2000	0.5261 (3.4275)	1.5982 (15.7291)
Financial Distress	Returns of highest minus lowest decile portfolios of all NYSE, AMEX and NASDAQ stocks, but excluding financial firms, sorted by Altman's (1968) Z-scores (see Boguth, Carlson, Fisher, and Simutin (2011)). Firms in the lowest Z-score decile portfolio have a higher probability of bankruptcy.	Feb. 1969 - Dec. 2000	0.3762 (1.1592)	0.6486 (2.2164)
Net Stock Issues	Returns of highest minus lowest decile portfolios of all NYSE, AMEX and NASDAQ stocks, excluding financial firms and firms with negative book equity, sorted by net stock issues in previous year (see Fama and French (2008)).	Feb. 1969 - Dec. 2000	-0.7355 (-4.5644)	-1.1052 (-6.5661)
Profitability	Value-weighted returns of highest minus lowest tercile portfolios of all NYSE, AMEX and NASDAQ stocks, excluding financial firms and firms with negative book equity, sorted by return on equity (see Chen, Novy-Marx, and Zhang (2011)).	Jan. 1972 - Dec. 2000	0.8158 (5.1129)	-
Illiquidity	Equally-weighted returns of highest minus lowest decile portfolios of all NYSE, AMEX and NASDAQ stocks, but excluding financial firms, sorted by Amihud's (2002) illiquidity measure (see Boguth, Carlson, Fisher, and Simutin (2011)).	Feb. 1969 - Dec. 2000	-	0.4517 (2.6622)

Table 6 lists return anomalies examined in this study, the sample period as well as the average value-weighted and equally-weighted monthly return for those cases where existing asset pricing models cannot explain the return. Table 6 also reports t -statistics for the null hypothesis that the average monthly return is zero.

Table 7

Residual correlations between return anomalies**Panel A: Value-weighted returns**

	Without <i>PIV</i> as a factor			
With <i>PIV</i> as a factor	Earnings Surprises	Financial Distress	Net Stock Issues	Profitability
Earnings Surprises		0.1179 (0.0278)	-0.0843 (0.1164)	0.3181 (0.0000)
Financial Distress	-0.0043 (0.9364)		-0.3307 (0.0000)	0.6116 (0.0000)
Net Stock Issues	-0.0016 (0.9769)	-0.1168 (0.0222)		-0.3059 (0.0000)
Profitability	0.2574 (0.0000)	0.3718 (0.0000)	-0.0733 (0.1726)	

Panel B: Equally-weighted returns

	Without <i>PIV</i> as a factor			
With <i>PIV</i> as a factor	Earnings Surprises	Financial Distress	Net Stock Issues	Illiquidity
Earnings Surprises		0.0946 (0.0781)	-0.0818 (0.1279)	0.0796 (0.1382)
Financial Distress	-0.0935 (0.0814)		-0.7152 (0.0000)	0.5985 (0.0000)
Net Stock Issues	0.0511 (0.3419)	-0.4982 (0.0000)		-0.5812 (0.0000)
Illiquidity	-0.0400 (0.4568)	0.3559 (0.0000)	-0.3824 (0.0000)	

Table 7 reports pairwise correlations coefficients between residuals of different return anomalies when the asset pricing model includes the Fama French, a momentum, and a liquidity factor (above diagonal entries) as well as the proposed *PIV* factor (below diagonal entries). In each case, we report the corresponding *p*-value in parantheses below the correlation coefficient. We employ the maximum number of observations in each case (see Table 6 for the sample period of each return anomaly).

Table 8
Explaining return anomalies with *PIV*

Panel A: Value-weighted returns

	Earnings Surprises		Financial Distress		Net Stock Issues		Profitability	
<i>Constant</i>	0.2939 (2.2463)	0.1993 (1.4441)	0.9432 (2.2982)	0.2168 (0.6627)	-0.7975 (-4.7468)	-0.5994 (-4.1011)	0.9146 (4.3102)	0.6050 (4.1565)
<i>MKT</i>	0.0107 (0.3279)	0.0568 (1.3857)	-0.2042 (-1.6200)	0.1449 (1.7161)	0.1214 (1.9590)	0.0262 (0.7393)	-0.0127 (-0.4229)	0.1385 (2.5933)
<i>SMB</i>	0.0007 (0.0117)	0.1318 (2.1331)	-0.6183 (-2.4664)	0.4389 (2.4720)	0.3530 (3.0708)	0.0647 (0.7078)	-0.2300 (-3.3116)	0.1996 (1.5868)
<i>HML</i>	0.0897 (1.2638)	0.1104 (1.5730)	-0.4110 (-2.4376)	-0.2806 (-2.0796)	-0.0278 (-0.3180)	-0.0633 (-0.9689)	-0.1489 (-0.9986)	-0.0813 (-0.6017)
<i>MOM</i>	0.1995 (2.5400)	0.2433 (4.0954)	-0.3645 (-1.0681)	-0.0643 (-0.5328)	0.0631 (0.6583)	-0.0188 (-0.4023)	0.0009 (0.0048)	0.1445 (1.6306)
<i>LIQ</i>	-0.0593 (-0.7980)	-0.0679 (-0.9630)	0.1371 (1.8207)	0.0567 (0.7390)	-0.0471 (-0.9442)	-0.0252 (-0.5415)	-0.0804 (-1.3963)	-0.1083 (-2.5869)
<i>PIV</i>		-0.0993 (-5.5970)		-0.7802 (-14.7109)		0.2128 (6.5779)		-0.3252 (-8.8316)
Adj. R^2	0.061	0.093	0.115	0.443	0.221	0.346	0.059	0.406

Panel B: Equally-weighted returns

	Earnings Surprises		Financial Distress		Net Stock Issues		Illiquidity	
<i>Constant</i>	1.4157 (8.2118)	1.3523 (8.9137)	1.1130 (3.4848)	0.4245 (2.7078)	-0.9810 (-5.3872)	-0.6759 (-5.7144)	0.5596 (2.9819)	0.2654 (1.5035)
<i>MKT</i>	0.0034 (0.1067)	0.0344 (0.9389)	-0.1533 (-2.2051)	0.1776 (2.1512)	0.1352 (2.9862)	-0.0115 (-0.3557)	-0.2509 (-6.8083)	-0.1095 (-2.4874)
<i>SMB</i>	-0.0351 (-0.4757)	0.0529 (1.0366)	-0.5540 (-4.5336)	0.4480 (3.6042)	0.1741 (2.0399)	-0.2700 (-3.6682)	0.1099 (1.3915)	0.5381 (4.0039)
<i>HML</i>	0.0204 (0.2622)	0.0342 (0.4866)	-0.2135 (-2.9781)	-0.0899 (-0.6415)	-0.2978 (-3.9663)	-0.3526 (-4.3618)	0.0715 (0.8427)	0.1243 (1.0505)
<i>MOM</i>	0.1880 (3.3603)	0.2174 (4.9308)	-0.3876 (-1.2989)	-0.1030 (-1.1433)	-0.0083 (-0.0596)	-0.1344 (-2.4628)	-0.0053 (-0.0336)	0.1163 (1.4353)
<i>LIQ</i>	-0.0593 (-1.2479)	-0.0650 (-1.2494)	0.0839 (0.9349)	0.0077 (0.1410)	-0.1001 (-2.2161)	-0.0664 (-1.6040)	-0.0203 (-0.4767)	-0.0529 (-1.4497)
<i>PIV</i>		-0.0666 (-3.0075)		-0.7396 (-14.5241)		0.3277 (12.6021)		-0.3161 (-26.1049)
Adj. R^2	0.149	0.183	0.157	0.620	0.251	0.524	0.120	0.370

Table 8 reports time-series regressions of returns of zero investment, highest minus lowest, portfolios of NYSE, AMEX and NASDAQ stocks, onto asset pricing factors (market excess return (MKT), size (SMB), value (HML), momentum (MOM), liquidity (LIQ), and predicted IV (PIV)). Newey-West (with up to 12 lags) t -statistics are reported in parentheses. For each regression, we employ the maximum number of observations (see Table 6 for the sample period of each return anomaly).

Table 9
Idiosyncratic volatility estimated from quote-midpoint prices
January 1985 - December 2000

Panel A: Portfolios sorted on idiosyncratic volatility

Predicted idiosyncratic volatility			
Rank	Mean	Std Dev	FF-3 Alpha
1	1.30	4.19	-0.06 (-0.70)
2	1.36	4.6	-0.05 (-0.48)
3	1.47	5.29	0.05 (0.41)
4	1.26	6.43	-0.06 (-0.33)
5	0.68	7.52	-0.63 (-3.16)
5-1'	-0.62		-0.57 (-2.34)

Panel B: Portfolios sorted on predicted and residual idiosyncratic volatility

Predicted idiosyncratic volatility				Residual idiosyncratic volatility			
Rank	Mean	Std Dev	FF-3 Alpha	Rank	Mean	Std Dev	FF-3 Alpha
1	1.34	4.06	-0.02 (-0.18)	1	0.89	6.33	-0.50 (-2.78)
2	1.38	5.22	-0.01 (-0.13)	2	1.31	5.09	-0.13 (-0.98)
3	1.42	6.31	0.07 (0.58)	3	1.28	4.53	-0.14 (-1.37)
4	1.21	7.37	-0.18 (-0.95)	4	1.45	4.41	0.10 (0.92)
5	0.39	7.85	-0.95 (-6.17)	5	1.34	4.95	-0.04 (-0.32)
5-1'	-0.95		-0.93 (-4.80)	5-1'	0.45		0.54 (2.20)

Table 9 presents return statistics of portfolios formed on idiosyncratic volatility estimated from quote-midpoint prices (Panel A) as well as the corresponding predicted (based on one common component) and residual idiosyncratic volatility (Panel B). Every month, stocks are sorted into five portfolios based on the stock's (predicted or residual) idiosyncratic volatility during the previous month. For each portfolio (1 through 5) we report (in percentage terms) the mean monthly return, standard deviation, as well as alpha with respect to the Fama–French (1993) three factor model. 5-1 refers to a strategy that is long the portfolio of stocks with the highest idiosyncratic volatility in the previous month and short the portfolio of stocks with the lowest idiosyncratic volatility in the previous month. For each estimate of alpha, we also report the corresponding robust (Newey-West with four lags) *t*-statistic.

Table 10

Comparing PIV and PIV estimated from quote-midpoint prices.

January 1985 - December 2000

Panel A: Value-weighted returns

	Earnings Surprises		Financial Distress		Net Stock Issues		Profitability	
<i>Constant</i>	0.0653 (0.3959)	0.0046 (0.0286)	0.2120 (0.4140)	-0.3631 (-0.6560)	-0.5767 (-3.4955)	-0.3936 (-2.5838)	0.5017 (2.1805)	0.3141 (1.0960)
<i>MKT</i>	0.0457 (1.0634)	0.0692 (1.4726)	-0.0214 (-0.1822)	0.2120 (1.7880)	0.1198 (2.5559)	0.0481 (0.9731)	0.0962 (1.7722)	0.1754 (2.0392)
<i>SMB</i>	0.0807 (1.0708)	0.0814 (1.1488)	0.1055 (0.3081)	0.0582 (0.2341)	0.2840 (2.7608)	0.2853 (2.6977)	0.1112 (1.2522)	0.0796 (1.7176)
<i>HML</i>	0.0916 (1.8036)	0.0858 (1.7571)	-0.3851 (-1.1416)	-0.4373 (-1.8085)	0.2172 (2.2782)	0.2343 (2.3414)	0.0143 (0.1003)	-0.0021 (-0.0115)
<i>MOM</i>	0.1619 (2.5251)	0.1498 (2.4679)	-0.1299 (-1.1551)	-0.3042 (-1.7418)	0.0413 (0.7873)	0.0814 (1.7620)	0.0953 (1.1265)	0.0203 (0.2186)
<i>LIQ</i>	-0.0883 (-1.2871)	-0.0828 (-1.2729)	0.0732 (0.4968)	0.1431 (1.0929)	0.0682 (1.2799)	0.0504 (0.8758)	-0.1316 (-3.0887)	-0.1034 (-1.9480)
<i>PIV</i>	-0.0787 (-3.7663)		-0.9093 (-24.4329)		0.2475 (13.0797)		-0.3462 (-11.8881)	
<i>Mid-quotes PIV</i>		-0.1105 (-3.7765)		-1.1917 (-13.3895)		0.3425 (9.5663)		-0.4323 (-6.2968)
<i>Adj. R²</i>	0.047	0.044	0.597	0.523	0.513	0.495	0.505	0.394

Panel B: Equally-weighted returns

	Earnings Surprises		Financial Distress		Net Stock Issues		Illiquidity	
<i>Constant</i>	1.3021 (10.8664)	1.2526 (10.2044)	0.3870 (1.5098)	-0.1021 (-0.2294)	-0.7069 (-3.3768)	-0.4945 (-1.8686)	-0.0382 (-0.1839)	-0.2391 (-0.8105)
<i>MKT</i>	0.0363 (1.0027)	0.0547 (1.4763)	0.1237 (1.1647)	0.3273 (2.0463)	0.0300 (0.6985)	-0.0598 (-1.0177)	-0.1866 (-3.6949)	-0.1023 (-1.6920)
<i>SMB</i>	0.0912 (1.5383)	0.0965 (1.6173)	0.2190 (1.2763)	0.1515 (2.2108)	-0.0437 (-0.5793)	-0.0067 (-0.0822)	0.3265 (3.2481)	0.2948 (3.2180)
<i>HML</i>	0.1083 (2.1335)	0.1035 (2.1137)	-0.2487 (-2.6615)	-0.2920 (-1.5410)	-0.1238 (-1.4979)	-0.1053 (-0.8354)	0.1718 (1.5866)	0.1542 (1.2001)
<i>MOM</i>	0.1882 (4.9084)	0.1837 (4.7879)	-0.1100 (-1.0795)	-0.2889 (-3.0984)	-0.0845 (-1.5571)	0.0018 (0.0325)	0.0806 (1.2390)	0.0027 (0.0350)
<i>LIQ</i>	-0.0334 (-0.9335)	-0.0305 (-0.8655)	-0.0672 (-1.4825)	0.0012 (0.0163)	0.0007 (0.0147)	-0.0315 (-0.5913)	-0.0749 (-1.4708)	-0.0455 (-0.7642)
<i>PIV</i>	-0.0495 (-2.7829)		-0.8570 (-26.5046)		0.3955 (10.7979)		-0.3639 (-7.3435)	
<i>Mid-quotes PIV</i>		-0.0773 (-3.3982)		-1.0871 (-11.1886)		0.4926 (12.0699)		-0.4570 (-8.3557)
<i>Adj. R²</i>	0.148	0.155	0.741	0.612	0.675	0.568	0.622	0.533

Table 10 reports time-series regressions of returns of zero investment, highest minus lowest, portfolios of NYSE, AMEX and NASDAQ stocks, onto asset pricing factors (market excess return (MKT), size (SMB), value (HML), momentum (MOM), liquidity (LIQ), predicted IV (PIV) and PIV estimated from quote-midpoint prices. Newey-West (with up to 12 lags) *t*-statistics are reported in parentheses.

Table 11

GMM results excluding low credit quality firms and controlling for innovations in average variance and correlation

		1	2	3	4
C O E F F I C I E N T	MKT	-9.2885 (-5.3623)	-6.9401 (-7.7483)	-5.3312 (-6.2338)	-4.9013 (-2.0921)
	SMB	-5.6928 (-1.5052)	1.0896 (0.5796)	-0.8200 (-0.3033)	-17.2975 (-2.7440)
	HML	-4.0383 (-1.2769)	-1.5120 (-0.9088)	-5.8182 (-3.2192)	-6.7690 (-3.2069)
	MOM	-14.3438 (-7.1715)	-4.1686 (-3.1575)	-12.7440 (-6.1335)	-9.5280 (-4.2236)
	LIQ	1.7196 (0.7056)	-1.1523 (-0.8080)	-2.0693 (-1.0408)	-2.3806 (-1.0683)
	PIV	6.9964 (3.5799)	3.6738 (4.1035)	2.6740 (2.1172)	7.5674 (3.4561)
	PAV				-23.3397 (-2.4626)
	PAC				-39.4203 (-2.7845)
P R E M I U M	MKT	0.0185 (4.6769)	0.0104 (6.0625)	0.0064 (3.4417)	0.0077 (4.4673)
	SMB	-0.0052 (-1.6156)	-0.0098 (-5.8631)	-0.0048 (-2.7124)	-0.0024 (-1.5503)
	HML	-0.0014 (-0.5702)	0.0020 (1.8781)	0.0005 (0.3959)	-0.0004 (-0.2720)
	MOM	0.0206 (6.5791)	0.0029 (1.6682)	0.0190 (6.0447)	0.0133 (4.1318)
	LIQ	-0.0018 (-0.4277)	0.0032 (1.6118)	0.0021 (0.8201)	0.0021 (0.7767)
	PIV	-0.0151 (-2.2475)	-0.0206 (-5.9803)	-0.0087 (-2.3015)	-0.0066 (-1.9538)
	PAV				0.0017 (1.4100)
	PAC				0.0028 (4.0469)
	<i>J</i>	29.3573 [0.2070]	40.7584 [0.0177]	37.9532 [0.0351]	23.4834 [0.3749]
	<i>HJ</i>	0.3011 [0.0001]	0.2533 [0.0000]	0.1829 [0.0018]	0.1669 [0.1007]
	Sample Period	Jan. 1986 - Dec. 2000	Jan. 1986 - Dec. 2000	Feb. 1969 - Dec. 2000	Feb. 1969 - Dec. 2000

Table 11 presents results from iterative GMM estimation of different asset pricing models and different test assets. The sub-panels "Coefficient" and "Premium" list the possible factors (market (*MKT*), size (*SMB*), value (*HML*), predicted *IV* (*PIV*), momentum (*MOM*), liquidity (*LIQ*), PAV and PAC - the mimicking portfolios for innovations in average variance and average correlation) considered in a linear pricing kernel. *PIV* refers to the case in which *IV* is predicted using the first common *IV* component. Column 1 employs the 30 Fama-French industry portfolios over a shorter time period (Jan. 1986 to Dec. 2000). Column 2 employs the 30 industry portfolios when the industry portfolios exclude firms in the lowest credit rating tercile (as determined by Avramov et al. (2012)). Columns 3 and 4 employ the monthly excess returns of the 30 Fama-French industry portfolios between Feb. 1969 and Dec. 2000. We report the coefficient estimate for each factor in the pricing kernel as well as the corresponding risk premium associated with the factor. Robust (Newey-West (1987) with one lag) *t*-statistics are in parentheses below the estimated coefficients and premiums. We report Hansen's (1982) *J*-statistic and Hansen-Jagannathan (1997) distance (*HJ*). *p*-values are in brackets below these test-statistics.

Table 12

Explaining return anomalies controlling for innovations in average variance and in average correlation

Panel A: Value-weighted returns

	Earnings Surprises		Financial distress		Net Stock Issues		Profitability	
Constant	0.199 (1.444)	0.246 (1.759)	0.217 (0.663)	0.148 (0.375)	-0.599 (-4.101)	-0.600 (-3.838)	0.605 (4.156)	0.534 (3.411)
MKT	0.057 (1.386)	0.063 (1.374)	0.145 (1.716)	0.080 (0.684)	0.026 (0.739)	0.070 (1.361)	0.139 (2.593)	0.111 (2.350)
SMB	0.031 (2.133)	0.168 (3.100)	0.439 (2.472)	0.459 (2.843)	0.065 (0.708)	0.000 (-0.006)	0.200 (1.587)	0.167 (1.313)
HML	0.109 (1.573)	0.136 (1.636)	-0.281 (-2.080)	-0.255 (-1.839)	-0.063 (-0.969)	-0.114 (-1.905)	-0.081 (-0.602)	-0.099 (-0.963)
MOM	0.243 (4.095)	0.245 (4.187)	-0.064 (-0.533)	-0.069 (-0.584)	-0.019 (-0.402)	-0.018 (-0.436)	0.144 (1.631)	0.141 (1.548)
LIQ	-0.068 (-0.963)	-0.070 (-0.982)	0.057 (0.739)	0.067 (0.841)	-0.025 (-0.542)	-0.035 (-0.772)	-0.108 (-2.587)	-0.104 (-2.360)
PIV	-0.099 (-5.597)	-0.100 (-5.234)	-0.780 (-14.711)	-0.775 (-14.780)	0.213 (6.578)	0.211 (8.354)	-0.325 (-8.832)	-0.322 (-9.415)
PAV		0.166 (0.952)		0.036 (0.124)		-0.235 (-2.205)		-0.167 (-1.413)
PAC		0.135 (0.584)		0.557 (1.107)		-0.632 (-2.785)		0.034 (0.132)
Adj. R2	0.093	0.095	0.443	0.443	0.346	0.365	0.406	0.408

Panel B: Equally-weighted returns

	Earnings Surprises		Financial Distress		Net Stock Issues		Illiquidity	
Constant	1.352 (8.914)	1.380 (8.718)	0.425 (2.708)	0.180 (1.020)	-0.676 (-5.714)	-0.605 (-4.447)	0.265 (1.504)	0.075 (0.355)
MKT	0.034 (0.939)	0.040 (0.945)	0.178 (2.151)	0.089 (1.008)	-0.011 (-0.356)	-0.001 (-0.018)	-0.110 (-2.487)	-0.164 (-3.212)
SMB	0.053 (1.037)	0.071 (1.347)	0.448 (3.604)	0.311 (2.434)	-0.270 (-3.668)	-0.208 (-2.501)	0.538 (4.004)	0.411 (2.854)
HML	0.034 (0.487)	0.046 (0.617)	-0.090 (-0.642)	-0.161 (-1.196)	-0.353 (-4.362)	-0.315 (-3.775)	0.124 (1.050)	0.054 (0.546)
MOM	0.217 (4.931)	0.218 (4.899)	-0.103 (-1.143)	-0.118 (-1.218)	-0.134 (-2.463)	-0.130 (-2.331)	0.116 (1.435)	0.105 (1.234)
LIQ	-0.065 (-1.249)	-0.066 (-1.275)	0.008 (0.141)	0.013 (0.270)	-0.066 (-1.604)	-0.065 (-1.575)	-0.053 (-1.450)	-0.052 (-1.518)
PIV	-0.067 (-3.008)	-0.067 (-3.189)	-0.740 (-14.524)	-0.726 (-12.988)	0.328 (12.602)	0.324 (11.653)	-0.316 (-26.105)	-0.306 (-23.145)
PAV		0.086 (0.914)		-0.625 (-2.869)		0.261 (1.813)		-0.559 (-2.623)
PAC		0.047 (0.470)		-0.047 (-0.147)		0.228 (1.308)		-0.235 (-1.064)
Adj. R2	0.183	0.181	0.620	0.636	0.524	0.532	0.3705	0.410

Table 11 reports time-series regressions of returns of zero investment, highest minus lowest, portfolios of NYSE, AMEX and NASDAQ stocks, onto asset pricing factors (market excess return (MKT), size (SMB), value (HML), momentum (MOM), liquidity (LIQ), predicted IV (PIV), PAV and PAC - mimicking portfolios for innovations in average variance and in average correlation. Newey-West (with up to 12 lags) *t*-statistics are reported in parentheses. For each regression, we employ the maximum number of observations (see Table 6 for the sample period of each return anomaly).

Figure 1
First common component of log idiosyncratic volatility
 July 1963 - December 2000

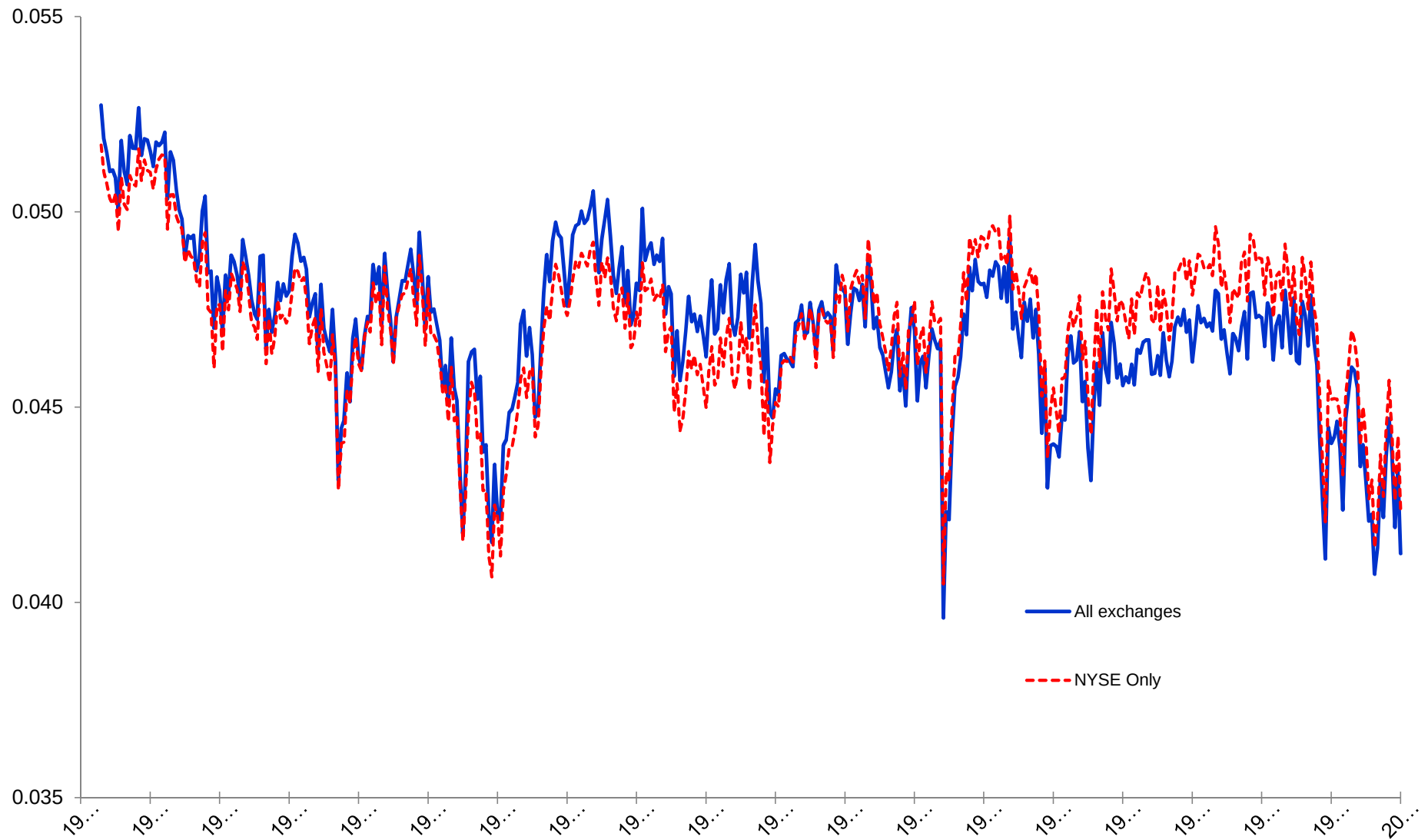
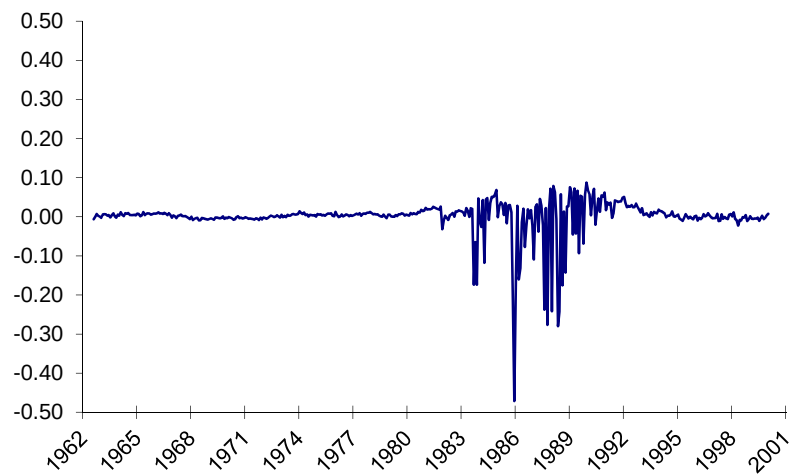


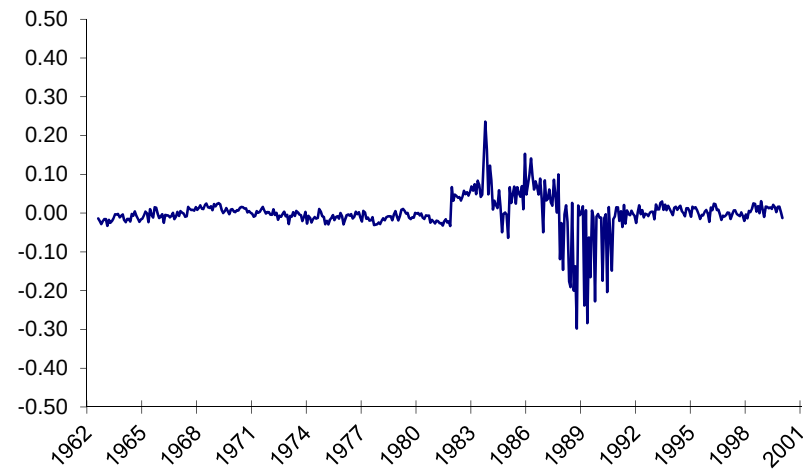
Figure 1 plots the time series of the first common component of log idiosyncratic return volatility for stocks listed on any of the three exchanges as well as for only those listed on the NYSE between July 1963 and December 2000.

Figure 2
Time series of common components 2-5: All exchanges
July 1963 - December 2000

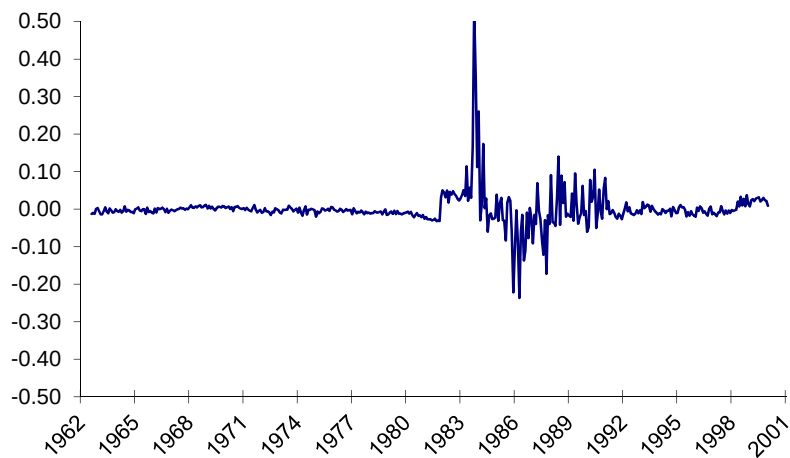
Panel A: Component 2



Panel B: Component 3



Panel C: Component 4



Panel D: Component 5

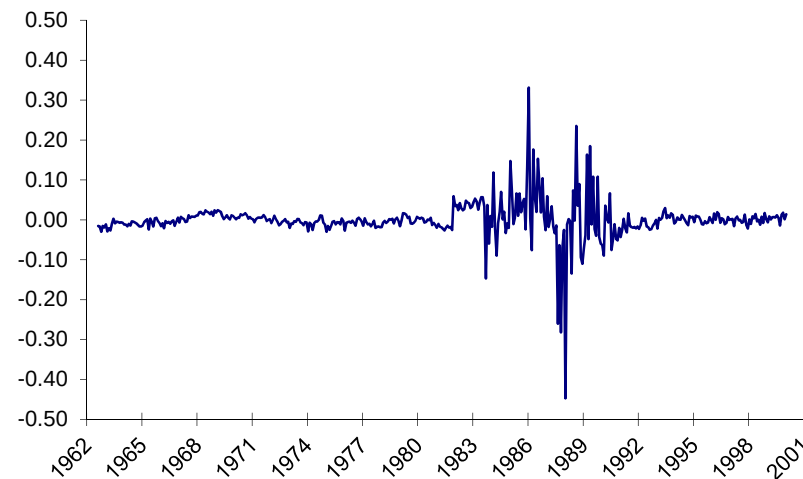
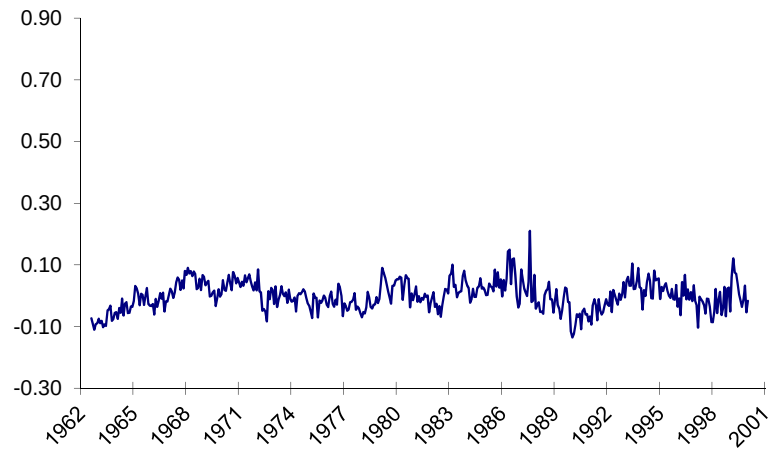


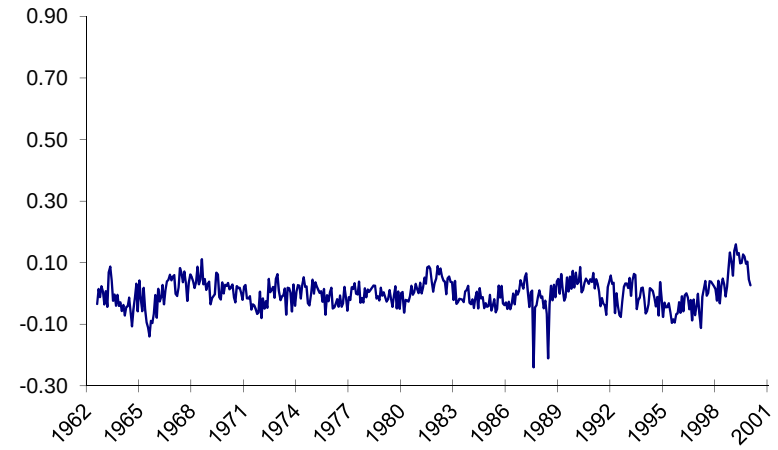
Figure 2 plots the time series of common components 2 through 5 of log idiosyncratic return volatility for stocks listed on any exchange between July 1963 and December 2000.

Figure 3
Time series of common components 2-5: NYSE only
July 1963 - December 2000

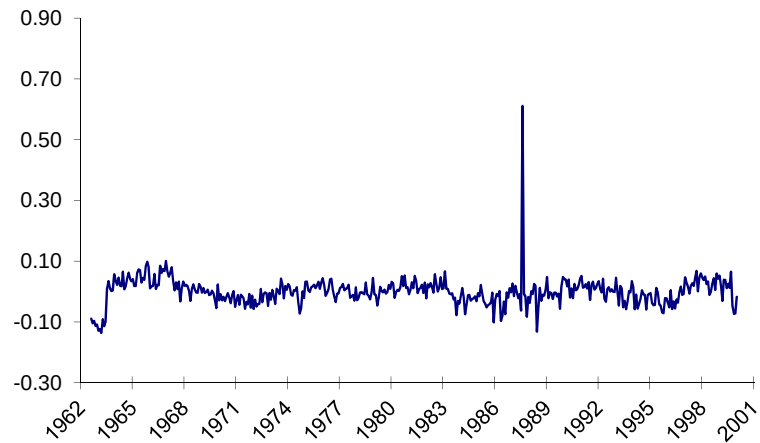
Panel A: Component 2



Panel B: Component 3



Panel C: Component 4



Panel D: Component 5

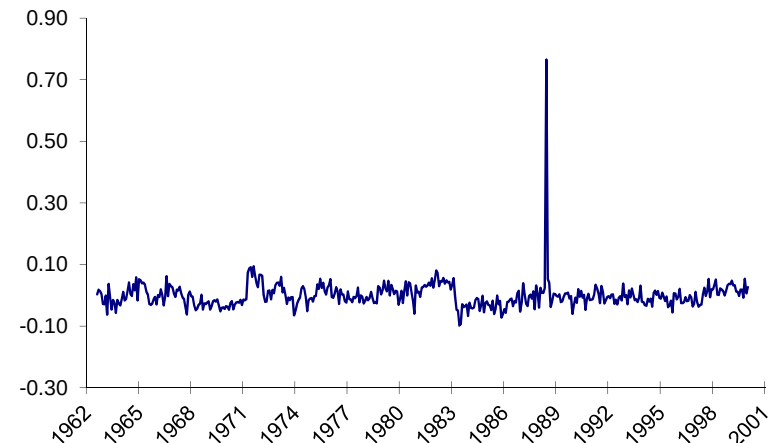


Figure 3 plots the time series of common components 2 through 5 of log idiosyncratic return volatility for stocks listed on the NYSE between July 1963 and December 2000.